

Question

X and **Y** are short-period elements of the same main group.

The reaction between the chloride of **X** and a compound containing approximately 65% of **Y** can prepare **A**; alternatively, the chloride of **X** and the hydride of **Y** in a 3:5 ratio can also yield compound **A** under high temperature. In **A**, the coordination number of **X** atoms is 4, while that of **Y** atoms is 2 and 3. Under high temperature and a nitrogen atmosphere, the reaction between **A** and salt **B**, or the 1:2 reaction between the lithium salt of **Y** and LiXY_2 , both produce compound **P**. The anion of **P** is isoelectronic with phosphorus pentoxide, while the anion of **B** is isoelectronic with the perchlorate ion.

Which of the following statements best fits the context?

- A.** The anion of **P** carries eight negative charges.
- B.** The anion of **P** possesses three fourfold rotation axes.
- C.** The equation for the preparation of **A** from the chloride of **X** and a compound containing approximately 65% of **Y**, with the coefficients in the simplest whole number ratio such that the sum of the coefficients of the reactants is greater than 20.
- D.** The lithium salt of **Y** reacts with LiXY_2 to form compound **P**. When the coefficients in the chemical equation for this reaction are in their simplest whole-number ratio, the sum of the coefficients of the reactants is 9.
- E.** It is known that above 1100K, **A** decomposes through a two-step reaction, each step producing an elemental substance, and the decomposition proceeds via the intermediate **O**; considering the chemical formula of **O** in its simplest integer ratio, its relative molecular mass is greater than or equal to 45.
- F.** The ratio of **Y** atoms with two-coordination and three-coordination in **A** is 2:3.

G. The chemical equation for the reaction between **A** and salt **B**, with the coefficients in the simplest whole-number ratio, has a sum of the product coefficients equal to 7.

H. None of the above options is correct.

Answer

Correct Answer: **G**

Detailed Explanation

X and **Y** are short-period elements in the same main group. Since **Y** forms a lithium salt, alkali metals and alkaline earth metals are first excluded.

CHECKPOINT

1 PTS

Y forms a lithium salt, excluding alkali metals and alkaline earth metals

The coordination numbers of the **Y** atom are 2 and 3, suggesting that **X** and **Y** are nitrogen and phosphorus.

CHECKPOINT

1 PTS

X and **Y** are inferred to be nitrogen and phosphorus

Since the compound LiXY_2 exists, **X** and **Y** cannot be chalcogens, halogens, or elements of Group 13, as no such compounds are known. Therefore, **X** and **Y** must belong to the nitrogen group, with **Y** having a lower coordination number, likely nitrogen, and **X** being phosphorus. In this case, LiXY_2 corresponds to LiPN_2 , a known compound, confirming the deduction.

CHECKPOINT

0.5 PTS

Y is nitrogen

CHECKPOINT

0.5 PTS

X is phosphorus

The chloride of **X** should be phosphorus trichloride or pentachloride. When reacting with a compound containing approximately 65% of **Y**, the mass fraction of nitrogen suggests the compound is sodium azide.

CHECKPOINT

1 PTS

The compound containing ~65% **Y** is sodium azide

The hydride of **Y** should be ammonia, which reacts with the chloride of **X** in a 5:3 ratio. Based on elemental knowledge and coordination number information, the product **A** is P_3N_5 , indicating the chloride is phosphorus pentachloride. The equations for the formation of **A** are:

CHECKPOINT

2 PTS

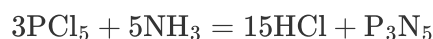
A is P_3N_5



The sum of the reactant coefficients is 18, making option C incorrect.

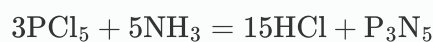
CHECKPOINT

1 PTS



CHECKPOINT

1 PTS



A is P_3N_5 , where each phosphorus atom has a coordination number of 4, while nitrogen atoms have coordination numbers of 2 and 3. Let the number of 2-coordinate nitrogen atoms be x ; the equation can be written as:

$$4 \times 3 = 2x + 3(5 - x)$$

Solving gives $x = 3$, so the ratio of 2-coordinate to 3-coordinate nitrogen is 3:2, making option F incorrect.

CHECKPOINT

1 PTS

In **A**, the ratio of 2-coordinate to 3-coordinate nitrogen is 3:2

P_3N_5 decomposes at high temperatures into two elemental substances, nitrogen and phosphorus. The initially formed product is nitrogen, so the intermediate, in its simplest ratio, is phosphorus nitride (PN). The combined relative molecular mass is 44.98, less than 45, making option E incorrect.

CHECKPOINT

1 PTS

The intermediate decomposition product, phosphorus nitride, has the simplest ratio 1:1, i.e., PN

The lithium salt of **Y** could be lithium nitride or lithium azide.

The lithium salt of **Y** reacts with LiPN_2 in a 1:2 ratio to form compound **P**. Since the anion of **P** is isoelectronic with P_2O_5 and likely contains nitrogen, and P_2O_5 typically exists as P_4O_{10} , the well-known anion structure $\text{P}_4\text{N}_{10}^{10-}$ is easily deduced. Thus, **P** is $\text{Li}_{10}\text{P}_4\text{N}_{10}$. The anion has an adamantane-like structure with a T_d point group, lacking a fourfold rotation axis, making options A and B incorrect.

CHECKPOINT

2 PTS

P is $\text{Li}_{10}\text{P}_4\text{N}_{10}$

CHECKPOINT

1 PTS

The $\text{P}_4\text{N}_{10}^{10-}$ anion belongs to the T_d point group and lacks a fourfold rotation axis

The reaction between the lithium salt of **Y** and LiPN_2 to form $\text{Li}_{10}\text{P}_4\text{N}_{10}$ suggests the lithium salt is lithium nitride. The equation is:



CHECKPOINT

1 PTS



The sum of reactant coefficients is 6, making option D incorrect.

P_3N_5 can also react with salt **B** to form $\text{Li}_{10}\text{P}_4\text{N}_{10}$. Since the anion of **B** is isoelectronic with perchlorate, it is deduced to be PN_4^{7-} , making **B** Li_7PN_4 . The reaction equation is:

CHECKPOINT

2 PTS

Salt **B** is Li_7PN_4



CHECKPOINT

1 PTS



The sum of product coefficients is 7, making option G correct.

In conclusion, option G is correct.